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Lab 4

Task 1

The screenshot displays the pgAdmin 4 web interface. On the left, the 'Object Explorer' pane shows a tree structure of database objects. The 'lab4' database is selected, and its 'Views' folder is expanded. The main pane shows a SQL query editor with the following query:

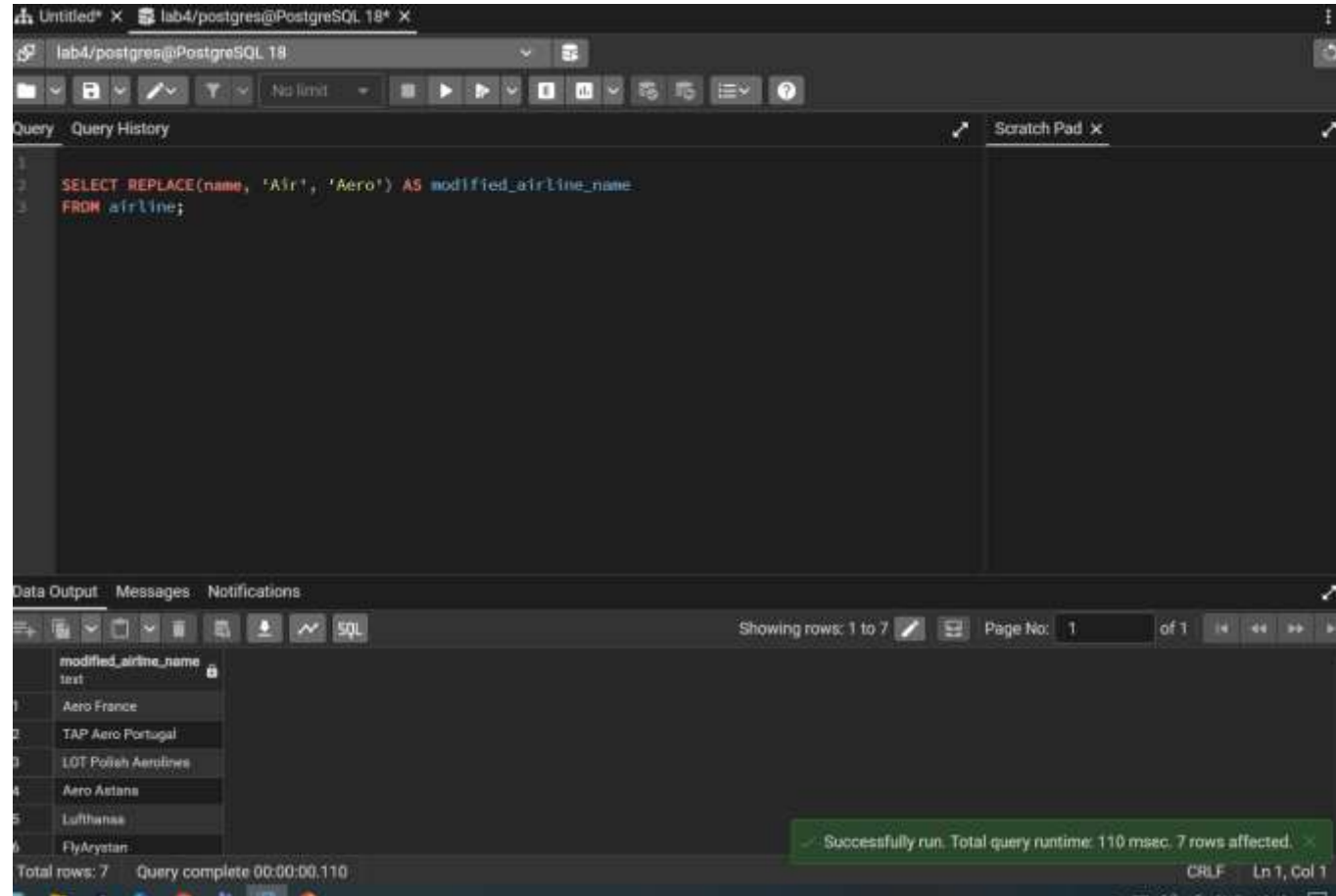
```
1  
2 SELECT UPPER(name) AS airline_name_uppercase  
3 FROM airline;
```

The 'Data Output' pane at the bottom shows the results of the query, displaying 7 rows of data. The first row is highlighted, showing the column 'airline_name_uppercase' with the value 'AIR FRANCE'.

airline_name_uppercase
AIR FRANCE
TAP AIR PORTUGAL
LOT POLISH AIRLINES
AIR ASTANA
LUFTHANSA
FLYARBYSTAN

A green status bar at the bottom right indicates: 'Successfully run. Total query runtime: 109 msec. 7 rows affected.'

Task 2



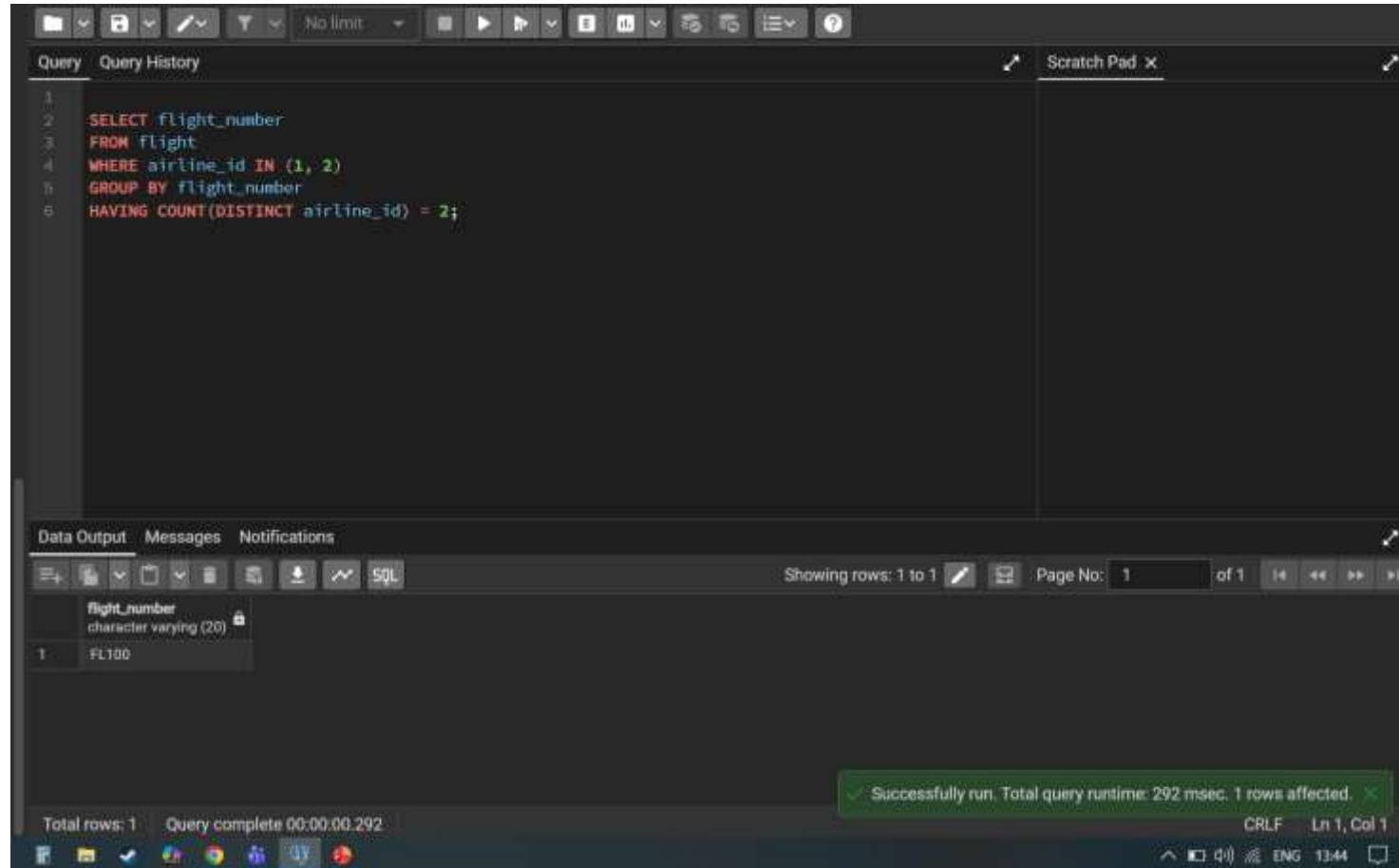
The screenshot shows a PostgreSQL client window with the following components:

- Query Editor:** Contains the SQL query:

```
1 SELECT REPLACE(name, 'Air', 'Aero') AS modified_airline_name
2 FROM airline;
```
- Data Output:** Displays the results of the query in a table format:

	modified_airline_name
1	Aero France
2	TAP Aero Portugal
3	LOT Polish Airlines
4	Aero Astana
5	Lufthansa
6	FlyArystan
- Messages:** A green notification bar at the bottom right states: "Successfully run. Total query runtime: 110 msec. 7 rows affected."
- Status Bar:** At the bottom, it shows "Total rows: 7", "Query complete 00:00:00.110", and "CRLF Ln 1, Col 1".

Task 3



The screenshot shows a SQL IDE interface with a dark theme. The top toolbar contains icons for file operations, query execution, and settings. The main editor area displays a SQL query:

```
1 SELECT flight_number
2 FROM flight
3 WHERE airline_id IN (1, 2)
4 GROUP BY flight_number
5 HAVING COUNT(DISTINCT airline_id) = 2;
```

Below the editor, the 'Data Output' tab is active, showing a table with one row:

flight_number
FL100

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.292'. A green notification box at the bottom right states: 'Successfully run. Total query runtime: 292 msec. 1 rows affected.'

Task 4

The screenshot shows a PostgreSQL query editor interface. The top bar indicates the connection is to 'lab4/postgres@PostgreSQL 18'. The query editor contains the following SQL code:

```
1.
2. SELECT *
3. FROM airport
4. WHERE name ILIKE '%Reginald%'
5. AND name ILIKE '%Air%';
```

The 'Data Output' tab is active, displaying the results of the query in a table format. The table has four columns: 'airport_id' (PK integer), 'name' (character varying (100)), 'city' (character varying (50)), and 'country' (character varying (50)). The results show one row with the following values:

airport_id	name	city	country
1	Reginal Air Gateway	City A	Country A

At the bottom of the interface, a green status bar indicates: 'Successfully run. Total query runtime: 195 msec. 1 rows affected.' The bottom status bar also shows 'Total rows: 1' and 'Query complete 00:00:00.195'.

Task 5

The screenshot shows a PostgreSQL client interface with a dark theme. The top bar displays the connection name 'lab4/postgres@PostgreSQL 18'. Below the toolbar, the 'Query' tab is active, showing a SQL query:

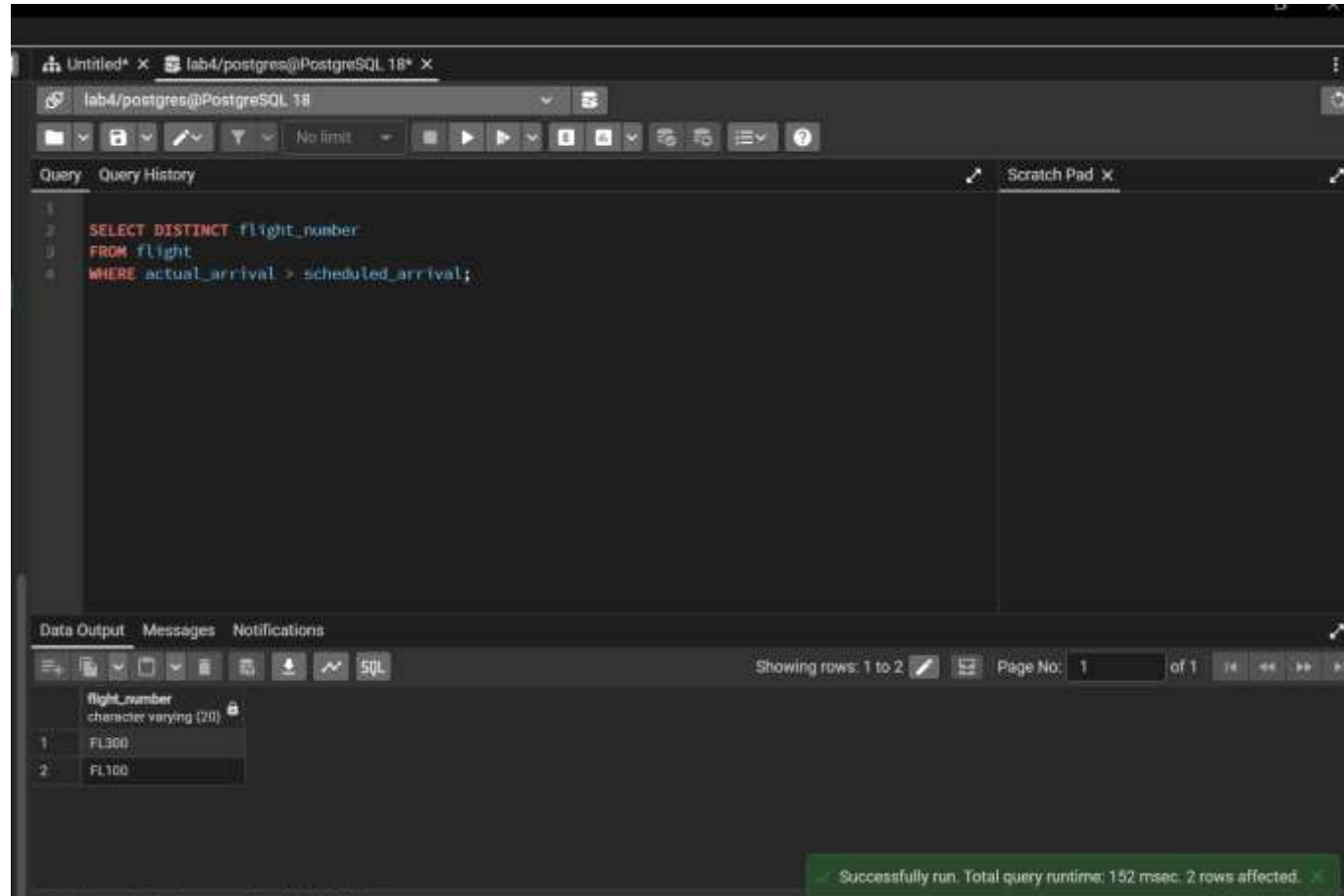
```
1  
2 SELECT  
3     name,  
4     TO_CHAR(birth_date, 'FMMonth DD, YYYY') AS formatted_birth_date  
5 FROM passenger;
```

The 'Data Output' tab at the bottom shows the results of the query in a table with two columns: 'name' (character varying (100)) and 'formatted_birth_date' (text). The table contains four rows of data:

	name	formatted_birth_date
1	John Doe	May 15, 1990
2	Jane Smith	October 20, 2005
3	Ali Khan	March 01, 2008
4	Elena Petrov	December 05, 1985

A green status bar at the bottom right indicates: 'Successfully run. Total query runtime: 203 msec. 4 rows affected.'

Task 6



The screenshot shows a PostgreSQL query editor interface. The top bar indicates the connection is to 'lab4/postgres@PostgreSQL 18'. The main editor area contains the following SQL query:

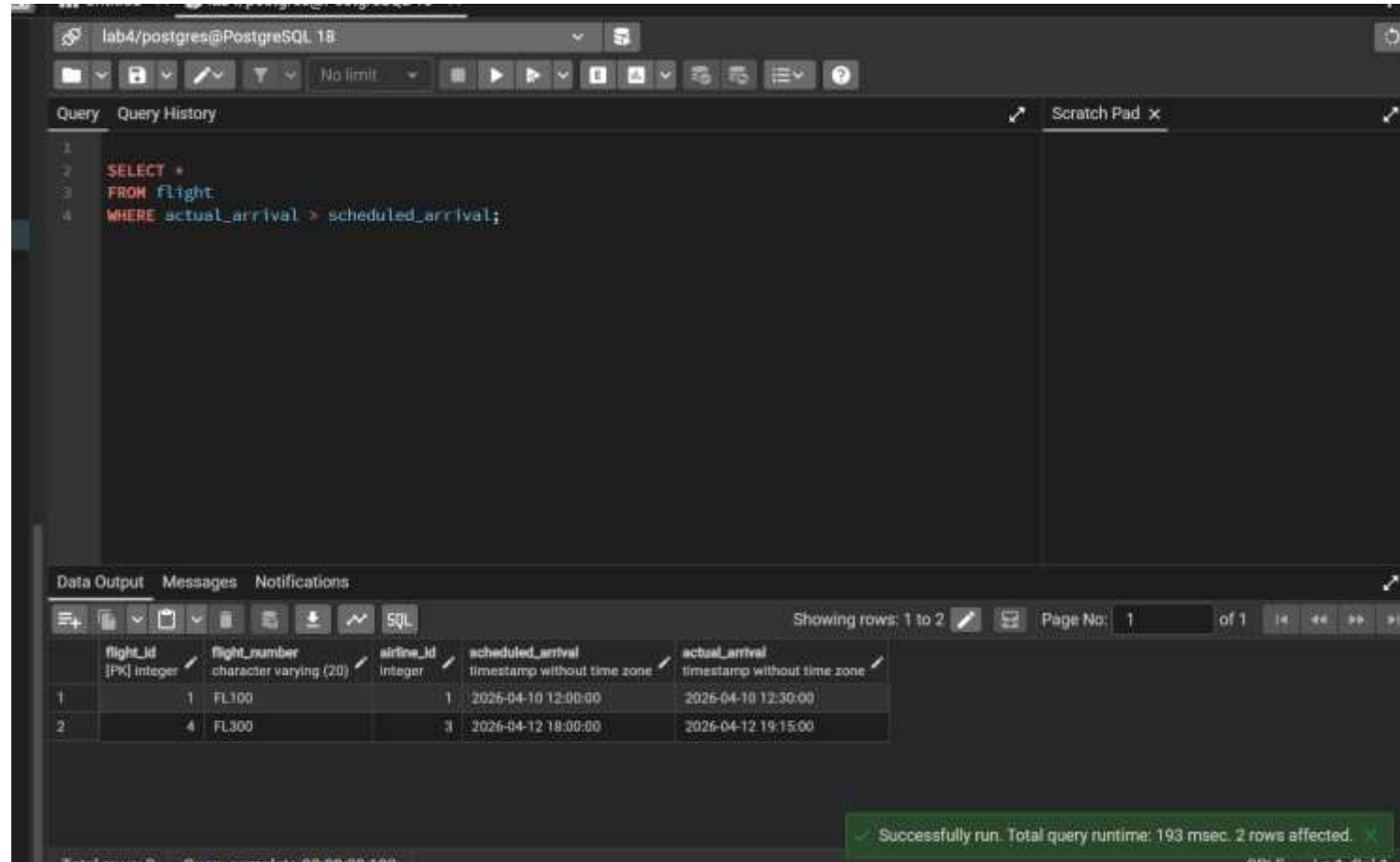
```
1  
2 SELECT DISTINCT flight_number  
3 FROM flight  
4 WHERE actual_arrival > scheduled_arrival;
```

Below the query editor, the 'Data Output' tab is active, displaying the results of the query. The results are shown in a table with the following structure:

	flight_number
1	FL300
2	FL100

The table has a column 'flight_number' of type 'character varying (20)'. The interface also shows a status bar at the bottom indicating 'Showing rows: 1 to 2' and 'Page No: 1 of 1'. A green message box at the bottom right states: 'Successfully run. Total query runtime: 152 msec. 2 rows affected.'

Task 7



The screenshot shows a PostgreSQL query editor interface. The query editor displays the following SQL query:

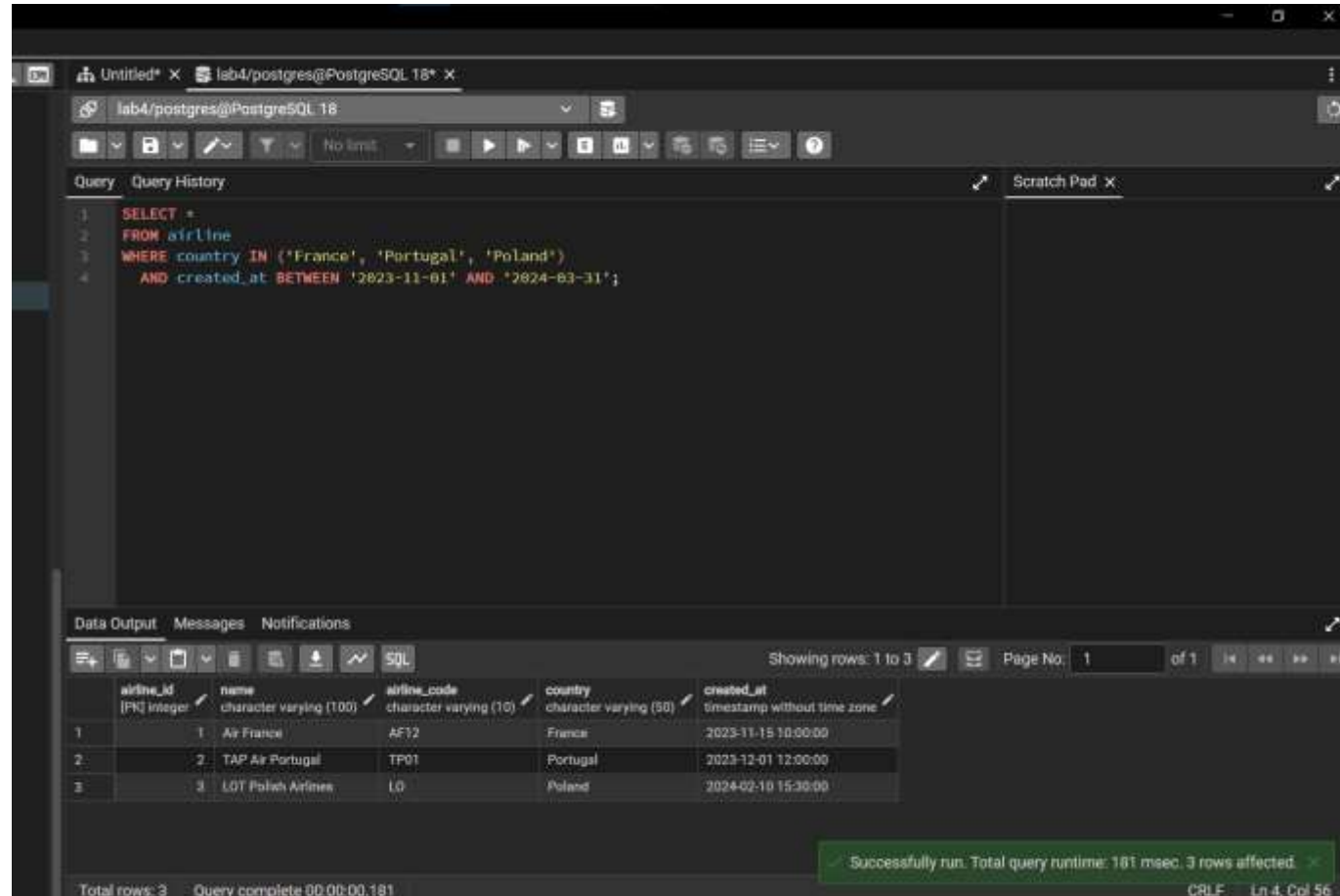
```
1  
2 SELECT *  
3 FROM flight  
4 WHERE actual_arrival > scheduled_arrival;
```

The query results are displayed in a table with the following columns: flight_id, flight_number, airline_id, scheduled_arrival, and actual_arrival. The results show two rows of data:

flight_id	flight_number	airline_id	scheduled_arrival	actual_arrival
1	FL100	1	2026-04-10 12:00:00	2026-04-10 12:30:00
2	FL300	3	2026-04-12 18:00:00	2026-04-12 19:15:00

A status message at the bottom indicates: "Successfully run. Total query runtime: 193 msec. 2 rows affected."

Task 8



The screenshot shows a PostgreSQL query editor interface. The query editor displays the following SQL query:

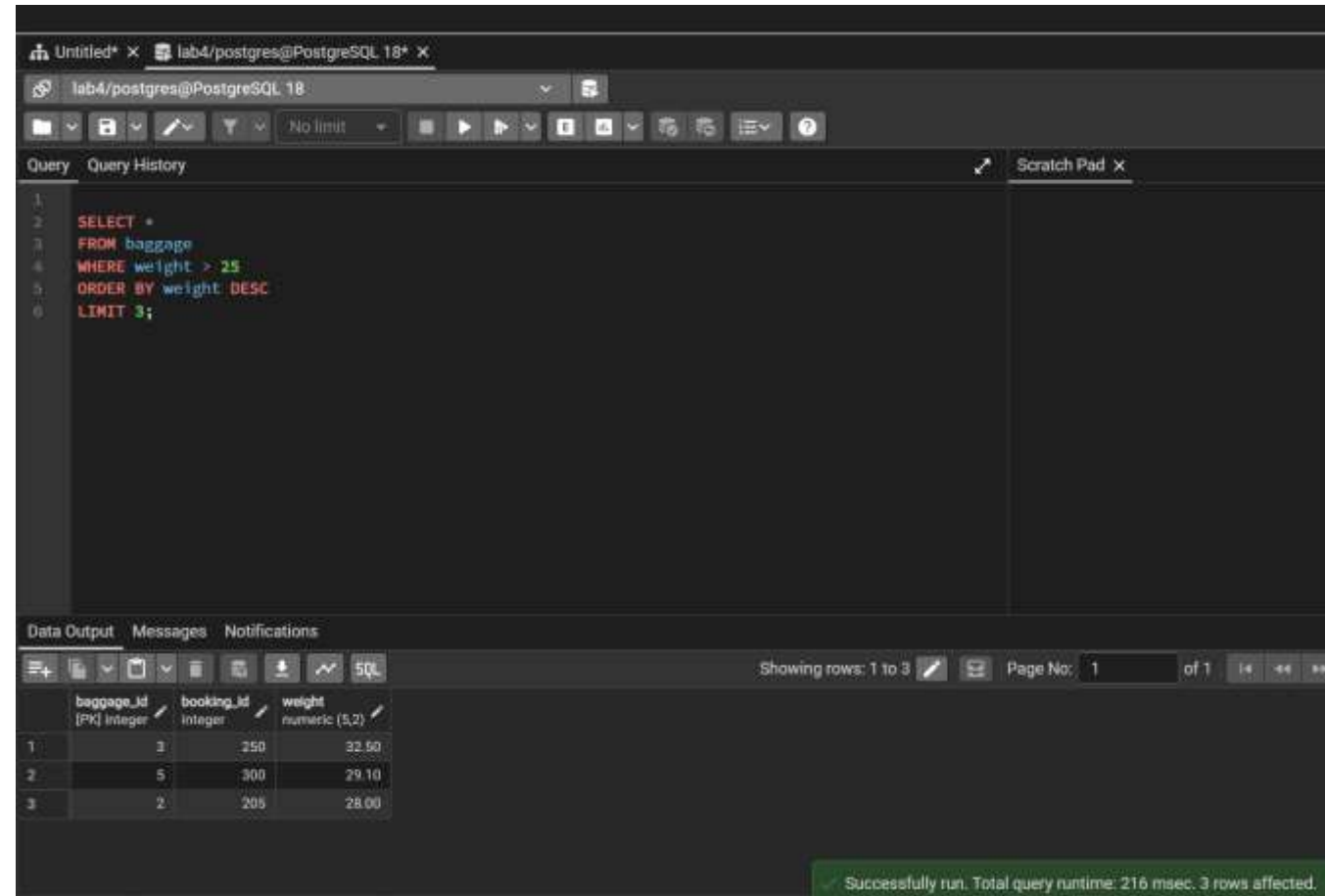
```
1 SELECT *
2 FROM airline
3 WHERE country IN ('France', 'Portugal', 'Poland')
4 AND created_at BETWEEN '2023-11-01' AND '2024-03-31';
```

The results are displayed in the Data Output tab, showing 3 rows of data:

airline_id	name	airline_code	country	created_at
1	Air France	AF12	France	2023-11-15 10:00:00
2	TAP Air Portugal	TP01	Portugal	2023-12-01 12:00:00
3	LOT Polish Airlines	LO	Poland	2024-02-10 15:30:00

A green status bar at the bottom indicates: "Successfully run. Total query runtime: 181 msec. 3 rows affected."

Task 9



The screenshot shows a PostgreSQL query editor interface. The top bar indicates the connection is to 'lab4/postgres@PostgreSQL 18'. The query editor contains the following SQL query:

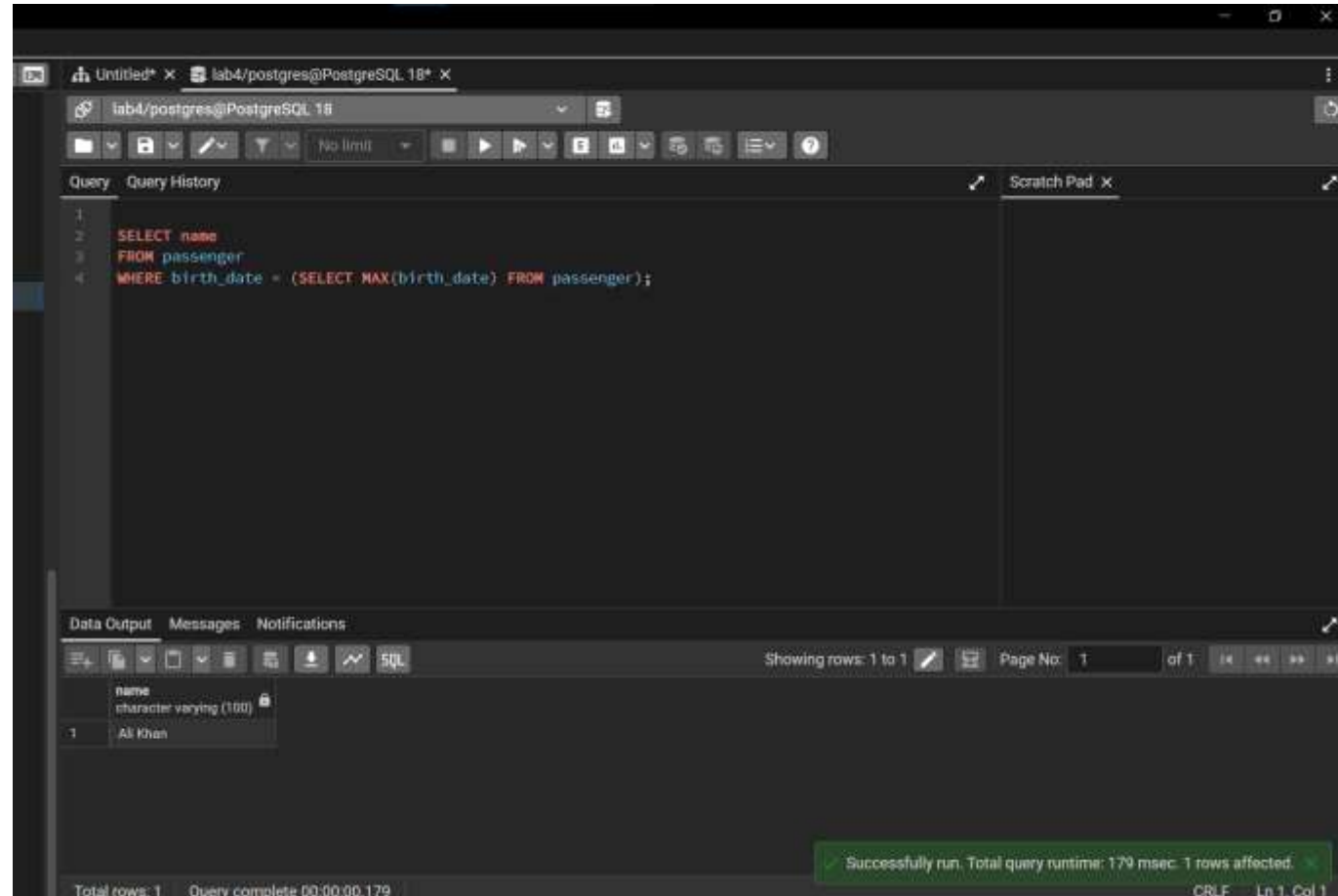
```
1 SELECT *
2 FROM baggage
3 WHERE weight > 25
4 ORDER BY weight DESC
5 LIMIT 3;
```

The results are displayed in a table with the following columns: baggage_id (PK) integer, booking_id integer, and weight numeric (5,2). The table shows 3 rows of data:

	baggage_id (PK) integer	booking_id integer	weight numeric (5,2)
1	3	250	32.50
2	5	300	29.10
3	2	205	28.00

At the bottom, a green status bar indicates: 'Successfully run. Total query runtime: 216 msec. 3 rows affected.'

Task 10



The screenshot shows a PostgreSQL query editor interface. The query editor has a toolbar with various icons for file operations, query execution, and formatting. The query text is as follows:

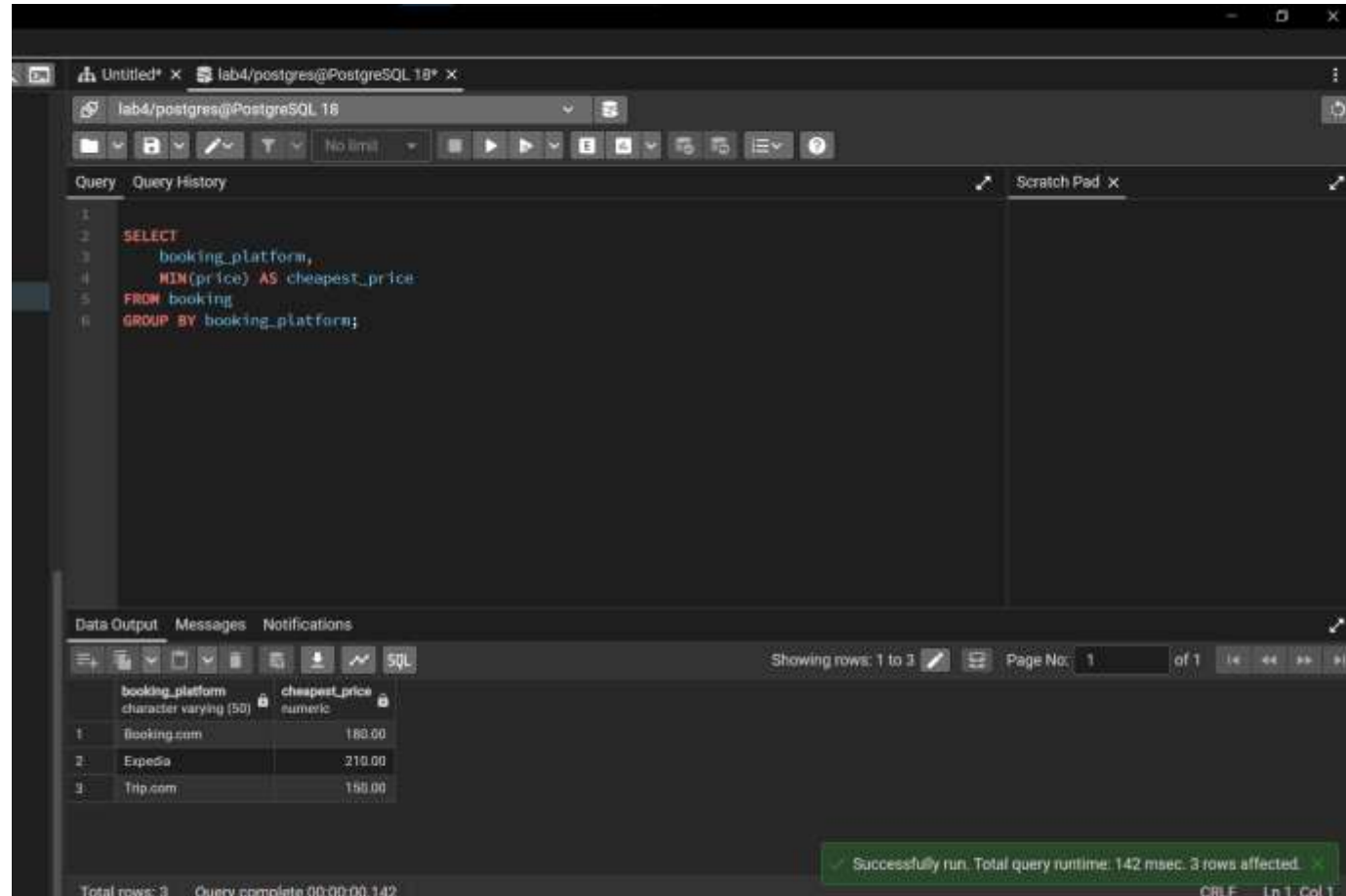
```
1  
2 SELECT name  
3 FROM passenger  
4 WHERE birth_date = (SELECT MAX(birth_date) FROM passenger);
```

Below the query editor, there is a "Data Output" tab showing the results of the query. The results are displayed in a table with one column, "name", and one row, "Ali Khan".

	name
1	Ali Khan

At the bottom of the interface, a status bar indicates "Total rows: 1" and "Query complete: 00:00:00.179". A green message box at the bottom right states: "Successfully run. Total query runtime: 179 msec. 1 rows affected."

Task 11



The screenshot shows a PostgreSQL query editor interface. The query editor displays the following SQL query:

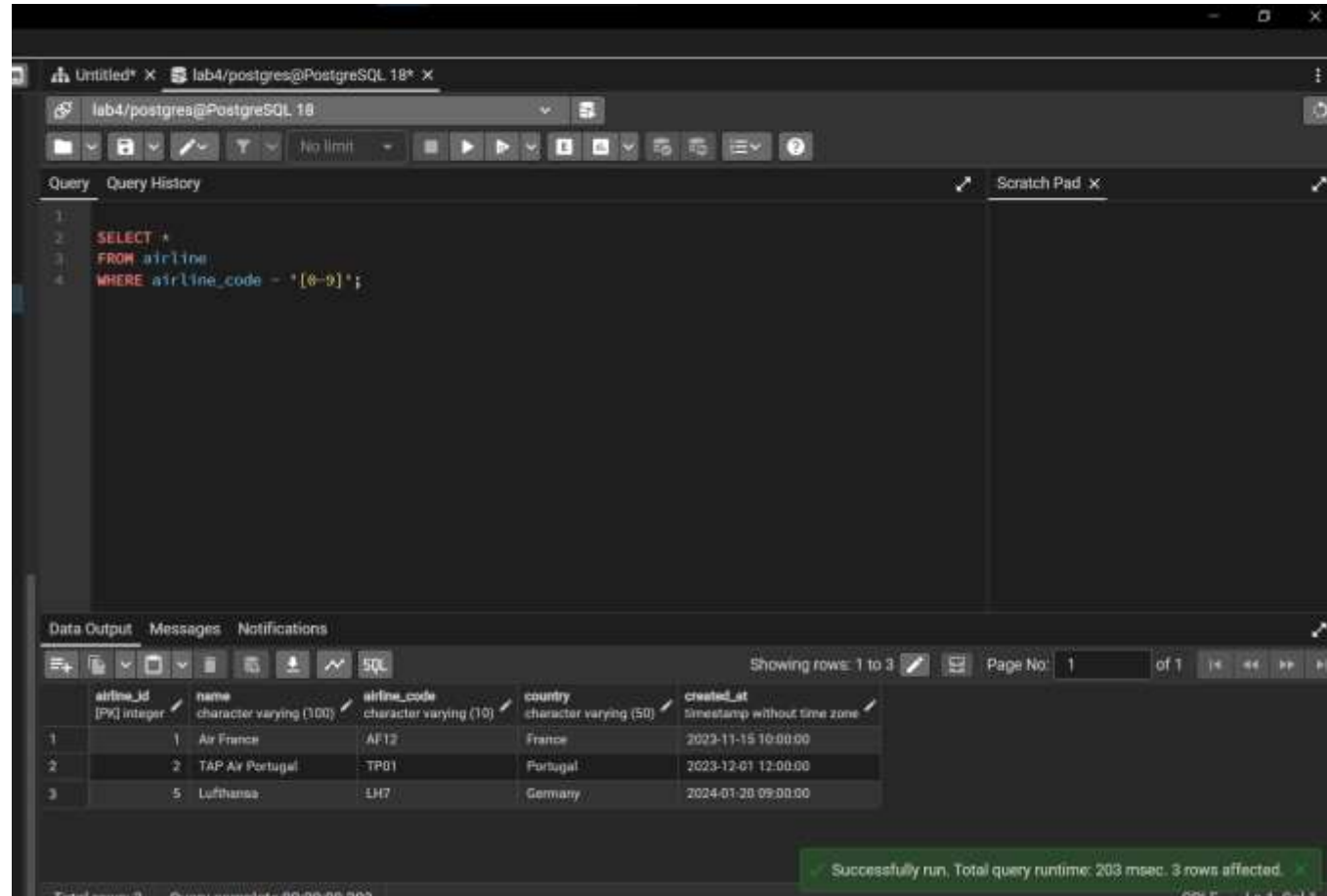
```
1  
2 SELECT  
3   booking_platform,  
4   MIN(price) AS cheapest_price  
5 FROM booking  
6 GROUP BY booking_platform;
```

The results are displayed in the Data Output tab, showing 3 rows of data:

	booking_platform character varying (50)	cheapest_price numeric
1	Booking.com	180.00
2	Expedia	210.00
3	Trip.com	150.00

A status message at the bottom indicates: "Successfully run. Total query runtime: 142 msec. 3 rows affected."

Task 12



The screenshot shows a PostgreSQL query editor interface. The query editor displays the following SQL query:

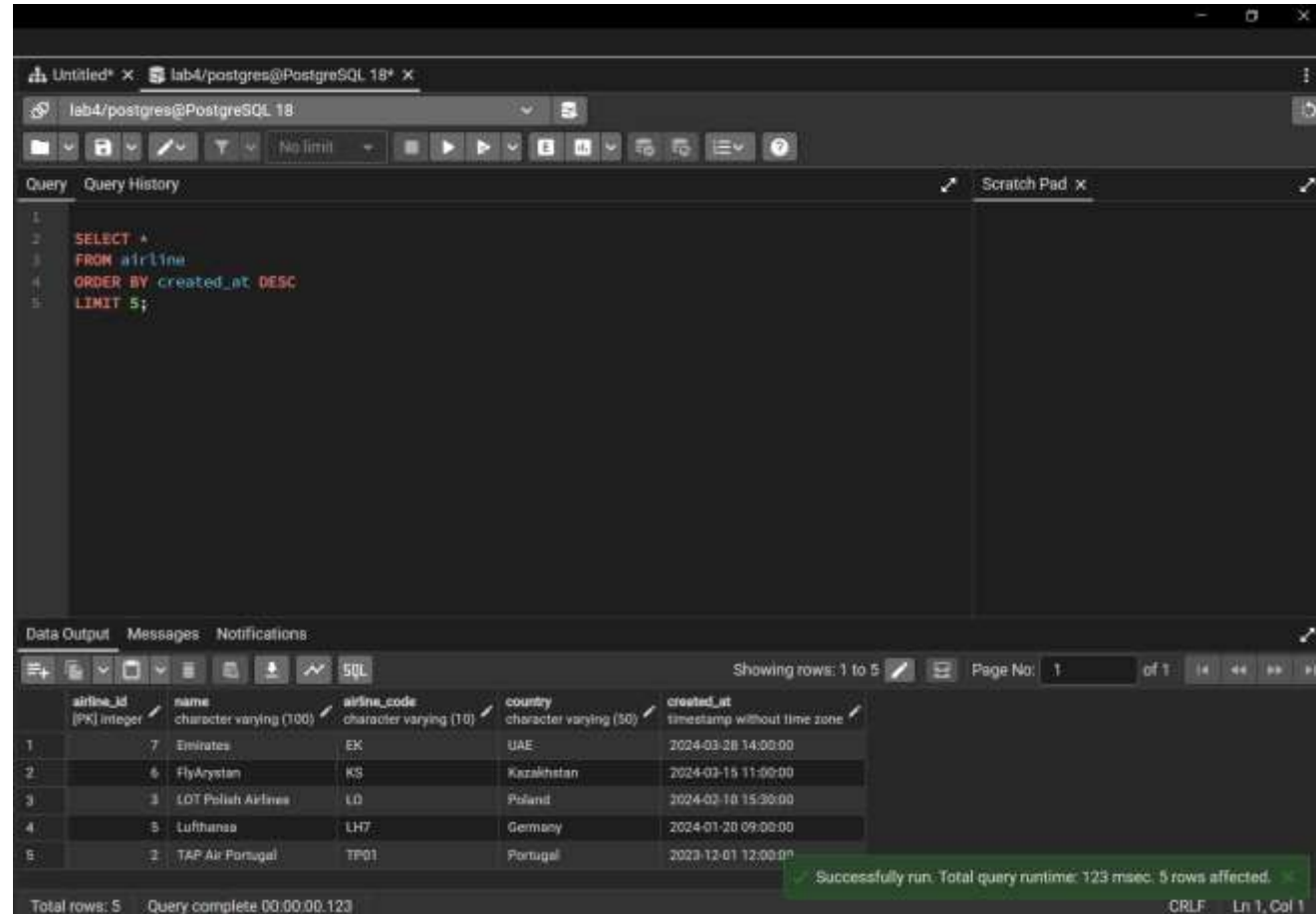
```
1  
2 SELECT *  
3 FROM airline  
4 WHERE airline_code ~ '[0-9]*';
```

The results are displayed in a table with the following columns: `airline_id` (integer), `name` (character varying (100)), `airline_code` (character varying (10)), `country` (character varying (50)), and `created_at` (timestamp without time zone). The table shows 3 rows of data.

	airline_id	name	airline_code	country	created_at
1	1	Air France	AF12	France	2023-11-15 10:00:00
2	2	TAP Air Portugal	TP01	Portugal	2023-12-01 12:00:00
3	5	Lufthansa	LH7	Germany	2024-01-20 09:00:00

At the bottom of the interface, a green status bar indicates: "Successfully run, Total query runtime: 203 msec. 3 rows affected."

Task 13



The screenshot shows a PostgreSQL query editor interface. The query editor displays the following SQL query:

```
1  
2 SELECT *  
3 FROM airline  
4 ORDER BY created_at DESC  
5 LIMIT 5;
```

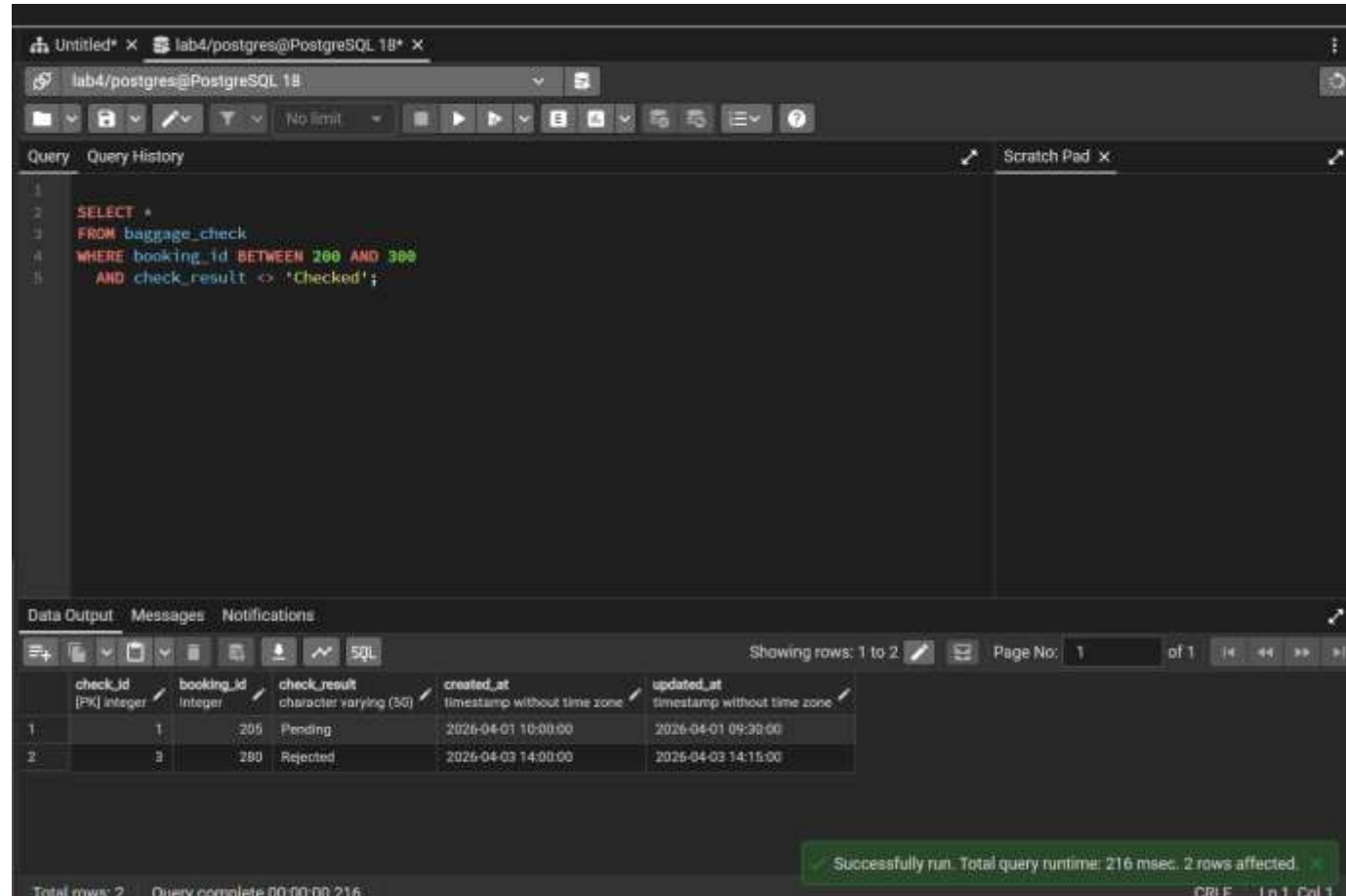
The results pane shows the following data output:

	airline_id [PK] integer	name character varying (100)	airline_code character varying (10)	country character varying (50)	created_at timestamp without time zone
1	7	Emirates	EK	UAE	2024-03-28 14:00:00
2	6	FlyArystan	KS	Kazakhstan	2024-03-15 11:00:00
3	3	LOT Polish Airlines	LO	Poland	2024-02-16 15:30:00
4	5	Lufthansa	LH7	Germany	2024-01-20 09:00:00
5	2	TAP Air Portugal	TP01	Portugal	2023-12-01 12:00:00

Successfully run. Total query runtime: 123 msec. 5 rows affected.

Total rows: 5 Query complete 00:00:00.123 CRLF Ln 1, Col 1

Task 14



The screenshot shows a PostgreSQL query editor interface. The top bar indicates the connection is 'lab4/postgres@PostgreSQL 18'. The query editor contains the following SQL code:

```
1  
2 SELECT *  
3 FROM baggage_check  
4 WHERE booking_id BETWEEN 200 AND 300  
5 AND check_result <> 'Checked';
```

The results are displayed in a table with the following columns: check_id (PK) Integer, booking_id Integer, check_result character varying (50), created_at timestamp without time zone, and updated_at timestamp without time zone. The table shows 2 rows of data.

	check_id (PK) Integer	booking_id Integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1	205	Pending	2026-04-01 10:00:00	2026-04-01 09:30:00
2		3	Rejected	2026-04-03 14:00:00	2026-04-03 14:15:00

A status message at the bottom indicates: 'Successfully run. Total query runtime: 216 msec. 2 rows affected.'

Task 15

The screenshot shows a PostgreSQL client window with the following components:

- Query Editor:** Contains a SQL query to select all rows from the `baggage_check` table where the month and year of `updated_at` match those of `created_at`, and `updated_at` is less than `created_at`.
- Data Output:** Displays the results of the query in a table format.
- Status Bar:** Shows the total number of rows affected and the query runtime.

SQL Query:

```
1 SELECT *
2 FROM baggage_check
3 WHERE EXTRACT(MONTH FROM updated_at) = EXTRACT(MONTH FROM created_at)
4 AND EXTRACT(YEAR FROM updated_at) = EXTRACT(YEAR FROM created_at)
5 AND updated_at < created_at;
```

Data Output Table:

	check_id [PK] integer	booking_id integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1	205	Pending	2026-04-01 10:00:00	2026-04-01 09:30:00

Status Bar: Total rows: 1. Query complete 00:00:00.220. Successfully run. Total query runtime: 220 msec. 1 rows affected.